The Hemoglobin Clinic: how to run the bench

A guide to both sessions, step by step. The pictures come from a run with the clinic's control DNA (a known HbA/HbS carrier, printed on its tube), not from any patient. Read the prelab review first: it explains each method.

The room and how to work in it

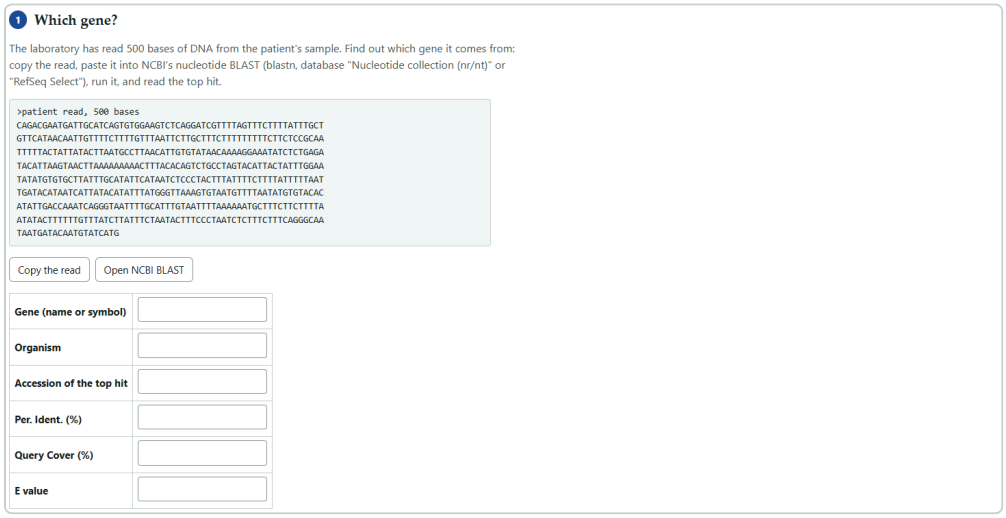


The room. Session 1 on the back bench, left to right: the computer, the PCR set-up, the thermocycler, the heat block and the sequencing drop-off. Session 2 on the right: the agarose, the gel box, the blue-light box.

- Choose a view with the buttons under the room (or click the room in the "Room" view).
- Work in the room itself, or with the controls under it and beside it. Every step has a control.
- A click on a pipette picks it up. Then a click on a tip box puts on a tip, on a stock draws from it, on a tube dispenses into it, on the orange bin ejects the tip. Places the pipette in hand could go next have a gold outline.
- If a step is about to go wrong, the bench warns you and lets you go back. If something has gone wrong, it says what, and you can start that part again.
- The bench clock runs on its own; "Speed up the wait" makes it run 60 times faster.

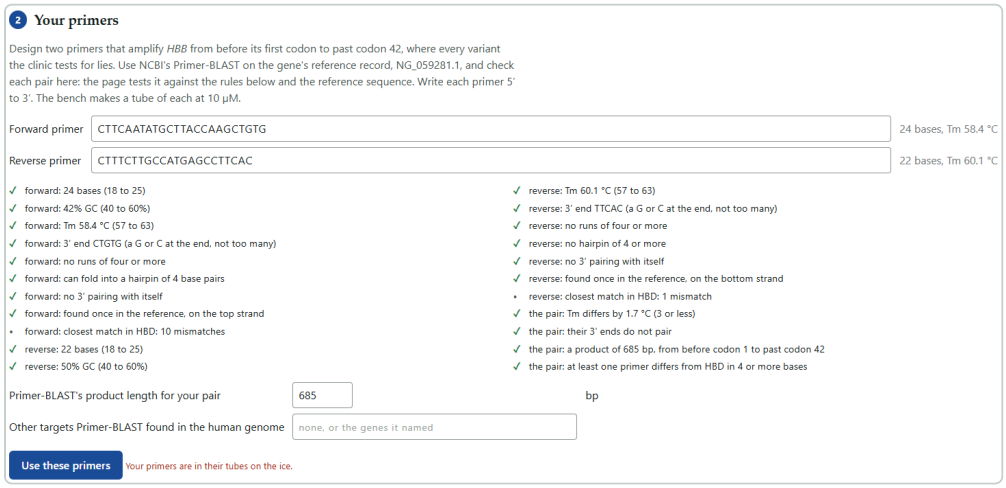
Session 1

1. Which gene?



Copy the read, open NCBI BLAST (blastn), paste it and run. Fill in the top hit's gene, organism, accession, identity, query cover and E value.

2. Your primers



Type your pair. The page checks each primer against the rules and against the reference sequence (a tick, or a "!" with what is wrong). Use NCBI Primer-BLAST on NG_059281.1, and give its product length and any other targets it found.

> The product must reach from before codon 1 to past codon 42. A pair that also matches HBD (delta-globin) copies it too: keep at least one primer where HBD differs. A primer's 3′ end on a base that differs between people stops it on that version.

3. The reaction, the digest, the sequencing

Work out every volume in the three recipe tables (25 µL PCR, 20 µL digest, 10 µL sequencing tube) and predict DdeI's fragments. The page checks each box.

4. Fill the three PCR tubes



The P20 in hand with a 200 µL tip on. The stocks it could draw from have a gold outline.

1. Choose the pipette (P2: 0.2 to 2 µL, 10 µL tips; P20: 2 to 20 µL and P200: 20 to 200 µL, 200 µL tips) and set its volume.
2. Put on a tip, draw from the stock, dispense into the tube, eject the tip.
3. A fresh tip for every liquid. A tip that has touched DNA and goes into the water puts DNA in the water; the tube with no DNA then shows a band.



The ice bucket: the six stocks at the back, the three PCR tubes in their cold block at the front (the Patient tube's cap is open).

The table beside the room shows what is in each tube. When all three hold 25 µL: cap each, flick it to mix, and give them a quick spin.

5. The thermocycler

A. Fill the three tubes

Set the pipette to the volume, put on a tip, draw from the stock and dispense into the tube (the controls are under the room). A tip that has touched one liquid carries a trace of it: use a fresh one each time it would go into something else.

In the tube (µL)	Patient	Control	No DNA
Water	9.5	9.5	10.5
Master mix	12.5	12.5	12.5
Forward primer	1	1	1
Reverse primer	1	1	1
DNA	1	1	0
Total	25	25	25
Now	capped, mixed, spun, in the thermocycler	capped, mixed, spun, in the thermocycler	capped, mixed, spun, in the thermocycler

B. Cap, mix, spin

1. **Cap each tube and flick it to mix**

   Patient: Cap | Uncap | Flick

   Control: Cap | Uncap | Flick

   No DNA: Cap | Uncap | Flick

2. **A quick spin**

   Spin the tubes

C. The thermocycler

1. **Put the tubes in and close the lid**

   Put all three in | Take all three out

   Open or close the lid

2. **The program**

Initial denaturation	95	°C	120	s
Cycles				
Denature	95	°C	30	s
Anneal		°C	30	s
Extend	72	°C		s
Final extension	72	°C	300	s

Send the program | **Run**

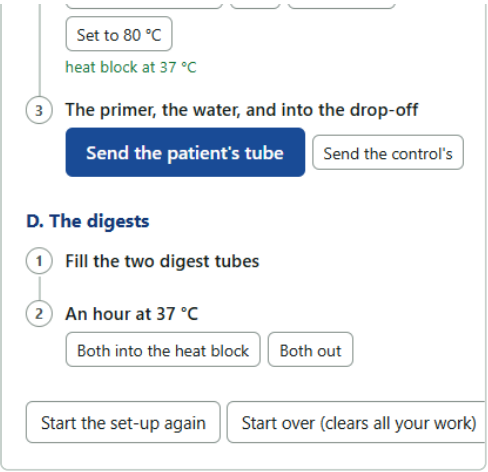
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Status                  Heating
Block                   32.1 °C
Cycle                   0 of 30
Time left               1:16:04
```

E. The tubes for sequencing

1. **Product and clean-up into "Sequencing: patient" (and the control, if you like)**

2. **15 minutes at 37 °C, then 15 at 80 °C**

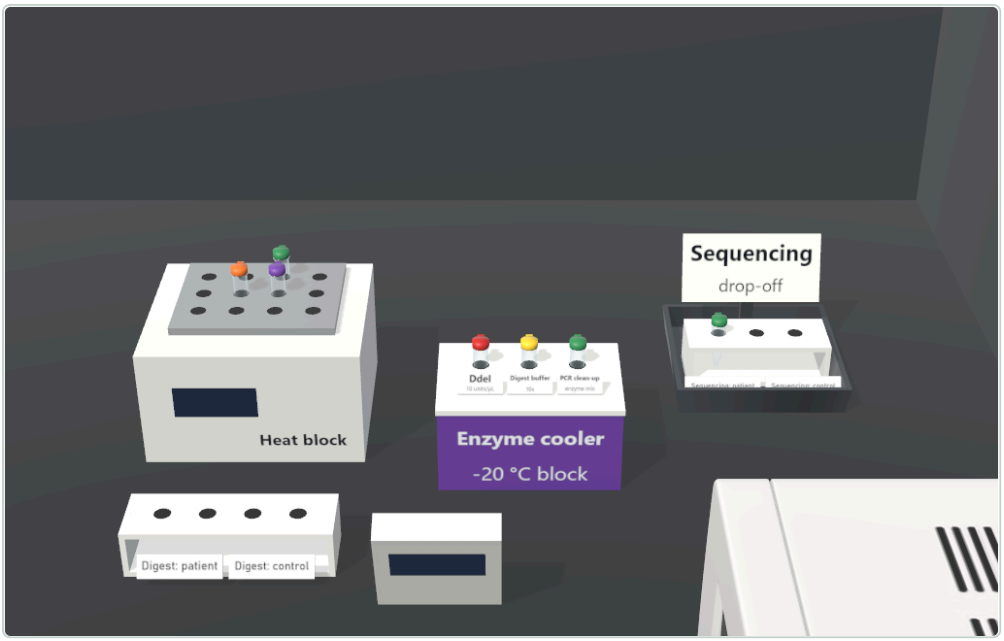
   Into the heat block | Out | Set to 37 °C



The controls beside the room, with the thermocycler's program and display.

1. Put the three tubes in and send the program: annealing a few degrees below your primers' lower Tm, an extension of about a minute for each 1000 bases, 30 cycles.
2. Close the lid and press Run. The run takes about 75 minutes of bench time. Do not open the lid until it is done.

6. The digests and the tube for sequencing



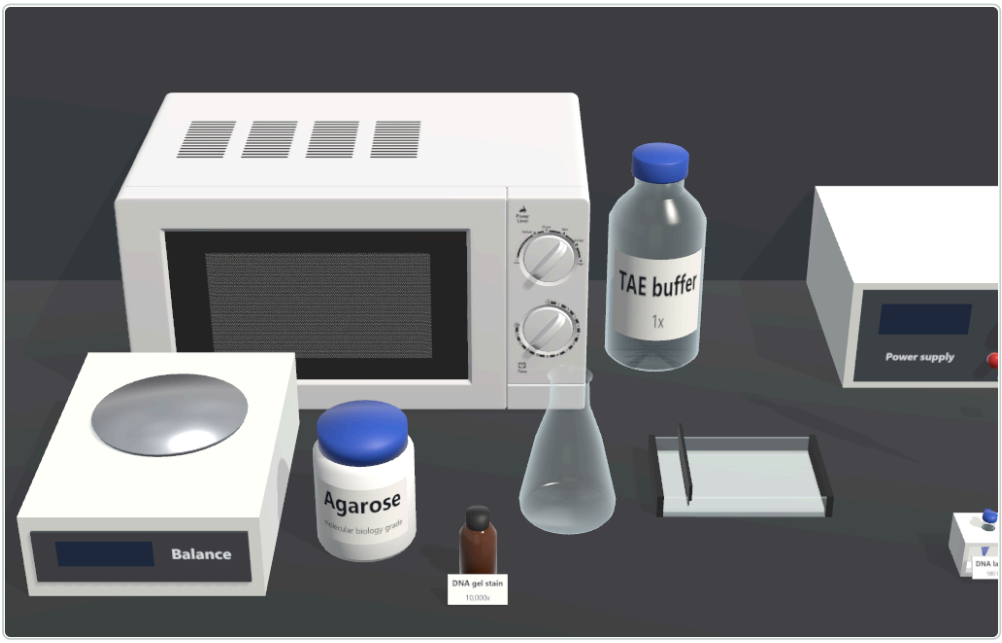
The digests and the sequencing tube in the heat block; the enzyme cooler with DdeI, its buffer and the clean-up enzymes; the sequencing drop-off.

1. Open the lid, take the tubes out and uncap them.
2. Each digest: 10 µL of PCR product, buffer, water, then DdeI last. An hour in the heat block at 37 °C.
3. The sequencing tube: product and clean-up enzymes; 15 minutes at 37 °C. Take the digests out, set the block to 80 °C for 15 minutes (this kills the clean-up enzymes, and would kill DdeI too).
4. Only then add one primer (forward or reverse) and the water, and send the tube.

> The clean-up enzymes eat single-stranded DNA. A primer added before the 80 °C step is eaten, and the trace comes back empty.

Session 2

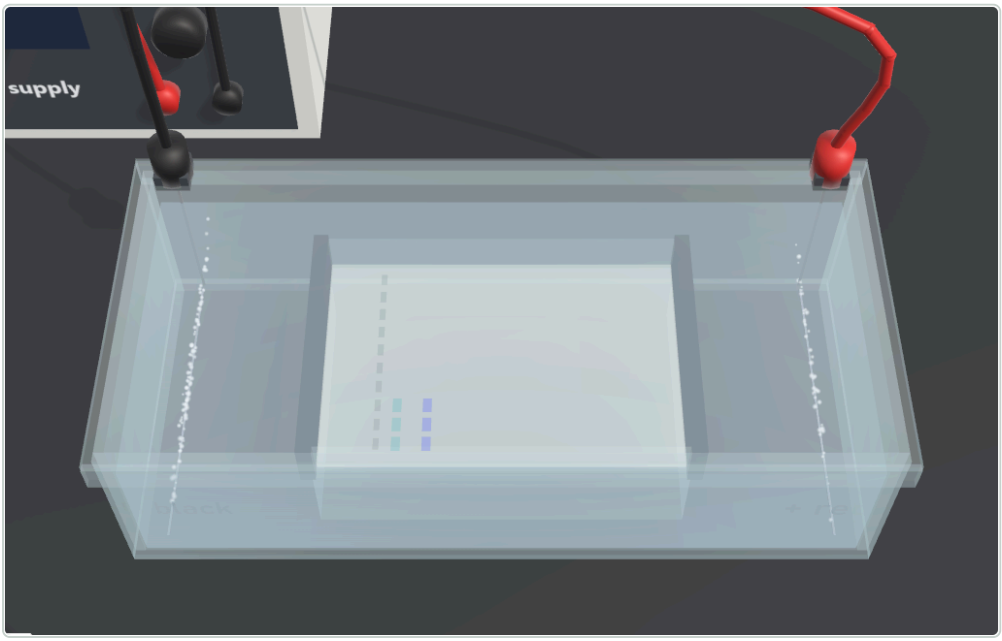
7. Cast the gel



The balance and agarose, the TAE, the flask, the microwave, the stain and the casting tray with its comb.

1. Choose the gel's strength and work out the agarose for 50 mL. Weigh it into the flask and add the TAE.
2. Melt it in short bursts in the microwave (30 s), swirling between them, until no grains float. A long burst boils it over.
3. Let it cool to about 55 °C. Add the stain with a pipette (1 µL for every 10 mL), set the comb, pour. Wait 20 minutes for it to set, then take out the comb.

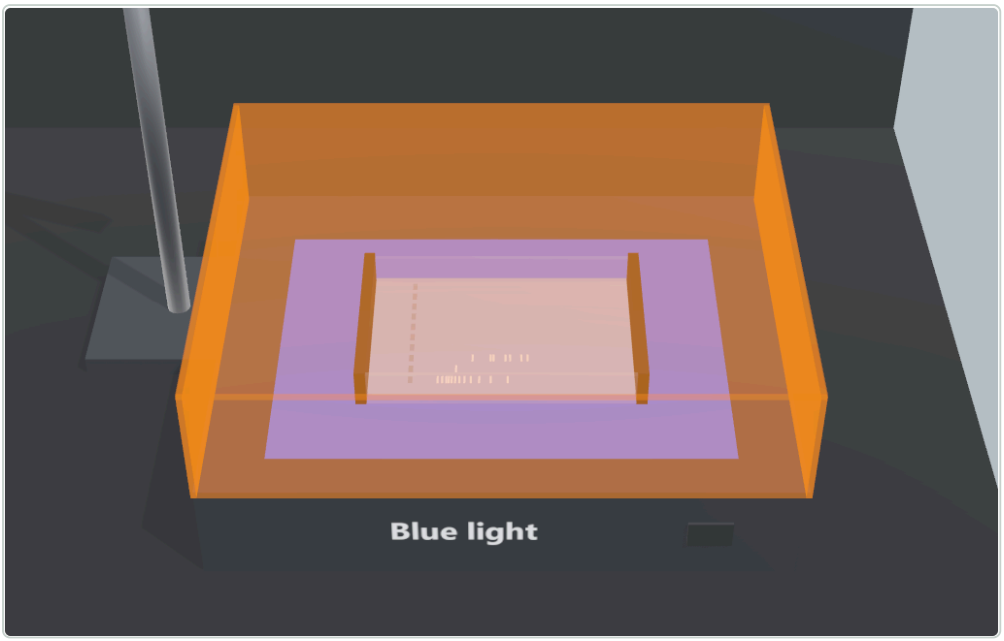
8. Load and run



The gel running: the leads on, bubbles at both electrodes, and the loading dye's two colours moving down the lanes.

1. The gel into the box with its wells at the black (−) end, then TAE to cover it.
2. Loading dye into each tube you will load (1 µL for every 5 µL of sample); the ladder comes ready to load.
3. Each sample into its own well, with a fresh tip. Keep a note of what goes where.
4. The lid on, the leads red to red and black to black, 100 V, power on. About an hour; stop before the faster dye reaches the end.

9. The picture



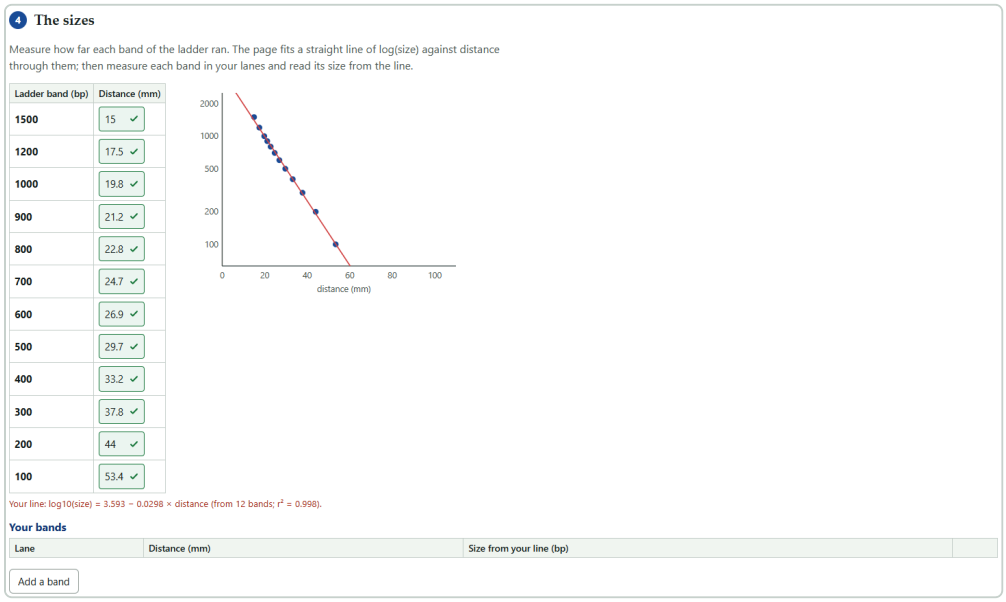
The gel on the blue-light box with the amber shield over it.

Power off; the gel onto the blue-light box; the light on; the amber shield over it; take the picture.



The picture: lane 1 the ladder, lane 2 the control's uncut product, lane 3 the control's digest. Measure from the bottom of the well to the middle of each band against the ruler.

10. The sizes



Type each ladder band's distance; the page draws the line of log(size) against distance. Then add a row for each band in your lanes and read its size from the line.

11. The sequence



The control's trace near position 349: a double peak of A (green) and T (red), called W.

1. Collect the results at the computer.
2. List each double peak (its position and its two bases). For each, the page asks where it lies in HBB: count from the ATG along the exons, and number the codons as Lehninger does (from the valine after the starting methionine).
3. No double peak? Compare the bases called with the reference: a change on both chromosomes shows no double peak.

12. The protein and the diagnosis

Give the normal β chain's first ten amino acids, look at HbA and HbS in 3D, then name each allele, the diagnosis, and the evidence. The page checks that your answer agrees with your own gel and trace; your instructor marks the diagnosis. Then print the report, or save it as a PDF, and upload it.